

E&I SERVICES PROJECTS

Test Report

Inclino Test

1. PROJECT INFORMATION

Project Number : CF2200160
 Project Name : Test for selection Inclinator
 Lead engineer : J Sterrenburg
 Engineer(s) : B van Leijenhorst
 Location : Papendrecht

2. TYPE OF TEST

- 1) The influence of speed and acceleration in all directions on a range of Inclination Sensors

3. DESCRIPTION OF TEST

All the sensors that have to be tested and the required reference sensors are mounted on a rigid base. This base is placed in a car and with the car a test track is driven. In chapter 11 are 2 pictures of the base in the Car. The test track has speed bumps, roundabouts, and Highway Junctions. All the data is logged during the test tracks.

4. USED EQUIPMENT

Reference sensors
 IXblue Octans3
 SBG Ekinox2-E
 Kongberg Seatex MRU 5+

Tested sensors
 SBG Ellipse-A-G5A4-B1 (3-Axis)
 Gemac IS2TK090-0-RL (2-Axis)
 Gemac Motus XC6MZ360-O (3-Axis)
 SignalQuest SQ-GIX-2021 (3-Axis)
 Posital AKS-090-2-CA01-HK2-PV (2-Axis)
 Posital AKS-360-1-CA01-VK2-PV (1- axis)
 Dis QG65D-KDXYh-090H-Can-CFM-UL (3-axis)
 Lord sensing MV5-AR-J1939 (3-Axis)
 ASC TSF-360-CAN-D2 (1- axis)
 IHC MEMS 010 (1- axis)
 Seatools Sinclino 200 (1- axis)

5. LOGGED DATA

Reference Sensors

IXblue Octans 3	serial	-Angle 2 Axis
SBG Ekinox	serial	-Angle 2 Axis
Kongberg Seatex MRU 5+	network UDP	-Angle speed and acceleration 3 axis

Tested sensors

SBG Ellipse	serial	-Angle speed and acceleration 3 axis
Gemac IS2TK090-0-RL	canbus	-Angle 2 Axis
Gemac Motus XC6MZ360-O	canbus	-Angle 2 Axis

SignalQuest SQ-GIX-2021	canbus	-Angle 2 Axis
Posital AKS-090-2-CA01-HK2-PV	canbus	-Angle 2 Axis
Posital AKS-360-1-CA01-VK2-PV	canbus	-Angle 1 Axis
Dis QG65D-KDXYh-090H-Can-CFM-UL	canbus	-Angle 2 Axis
Lord sensing MV5-AR-J1939	canbus	-Angle 2 Axis
ASC TSF-360-CAN-D2	canbus	-Angle 1 Axis
IHC MEMS 010	network UDP	-Angle 1 Axis
Seatools Sinclino 200	4-20mA	-Angle 1 Axis

6. ANY FURTHER ACTION REQUIRED?

After selection of possible Sensors, a static test is necessary.

Then also the remaining angle "Z" could be tested.

7. OBSERVATIONS

Linear accelerations on the highway

At deceleration (left side of the Graph) and accelerations (right side) a big overshoot >15 deg is noticed at the following sensors

Gemac IS2TK090-0-RL
ASC TSF-360-CAN-D2
IHC MEMS 010
Seatools Sinclino 200
See Graph 1

At the same deceleration (left side of the Graph) and acceleration (right side) the following Sensor also has some overshoot. <10 but >5

Posital AKS-090-2-CA01-HK2-PV
Posital AKS-360-1-CA01-VK2-PV
The signals are overlapping what was to expect.
See Graph 2

At Graph 3 result for the following Sensors are given

There is some overshoot <5 but >1.5
SBG Ellipse
Dis QG65D-KDXYh-090H-Can-CFM-UL

At Graph 4 result for the following Sensors are given

There is some overshoot < 2.5 but most of the signals are following the reference signal (octans)

Driving a Curve on a Highway Junction will give an acceleration sideways and will influence the Roll and the influence on the Pitch should be minimum.

At Graph 5 the pitch indication of the Posital sensors is given in relation to the Pitch of the Octans. A small influence is expected but is very high.

In Graph 6 the indication of the Pitch start to be more in line with octans after 10 seconds (start at 122843) and then start to follow the octans.

In Graph 7 most of the sensors are following the indication of the Octans best sensor to follow is the Lord followed by the Signal Quest and the SBG ellipse and Gemac Modus the Dis sensor give a big variation on the signal.

Driving on local road with minimum acceleration and deceleration.

In Graph 8 to 14 the results are compared with the Octans.

The Yellow line is the acceleration in the pitch direction.

The Gray line is the acceleration in the roll direction.

The Red line is the Pitch or Roll of the Octans

In Graph 8 the result of IHC Mems and Posital in relation to Octans and the different accelerations.

Here you see the acceleration has a big influence on the pitch. An acceleration of 1.5 m/s² results in a increase of 7deg in the pitch for the IHC sensor and 4 deg for the posital

In Graph 9 the result of ASC, Seatools and Gemac IS2TK09 in relation to Octans and the different accelerations. The readings of the ASC, Seatools and Gemac IS2TK09 are almost the same.

Here you see the acceleration has a big influence on the pitch. An acceleration of 1.5 m/s² results in a increase of 7deg in the pitch for the ASC, Seatools and Gemac IS2TK09 sensor.

In Graph 10 the result of Signal Quest, Dis and Gemac XC6MZ in relation to Octans and the different accelerations. The readings of Signal Quest, Dis and Gemac XC6MZ behave the same. Of the Three sensors the Dis is the most sensitive and the Signal Quest the least sensitive for acceleration.

In Graph 11 the result of SBG Ellipse, SBG EKINOX and Signal Quest in relation to Octans and the different accelerations. The readings of Sensors behave the same. The SBG Ellipse and the Signal Quest are behaving similar, but both are more sensitive than the SBG ekinox for acceleration.

In Graph 12 the result of Lord, SBG EKINOX and MRU in relation to Octans and the different accelerations. The readings of Sensors behave the same. The SBG Ekinox is more sensitive for acceleration then the Lord and MRU. In most cases the lord, mru and Octans behave the same.

As a bonus we also did a test how the Yaw angle of the SBG Ellipse and Gemac XC6MZ behave compared to the Octans heading.

The Yaw angles are corrected at the start to the Octans Heading.

Graph 13 is giving the heading of the different sensors.

After 4 min the difference between Octans and SBG ellipse start to become > 5 deg.

And after 9min the difference between Octans and Gemac XC6MC start to become .> 5 deg.

We also wanted to monitor the Yaw of the other sensors, but the Yaw angle was not available in this type of sensor.

For the Lord there is a Yaw angle available but only in the Canbus version and here we use the Lord with J1939 protocol.

At the same test we also logged the filtered Pitch and Roll of the Gemac XC6MC

In Graph 14 you see the Roll of Octans and Gemac XC6MC at the same part of road at different times.

Left side of the graph the Raw data on the right filtered data the roll of the Gemac XC6MC is behaving the same only on the right you see some more stable readings what is to be expected.

8. LIMITATIONS OF THE TEST

All the sensors are mounted on a different location on the rigid base.

Each sensor will therefore be under a different speed and or acceleration but will have the same Pitch and Roll if we drive over speedbumps, roundabouts and highway Junctions.

This test is only to determine what the influence is of the speed and acceleration on the pitch and or roll indication of the different Sensors.

9. MOUNTING OFFSETS

Nul Ref point MRU

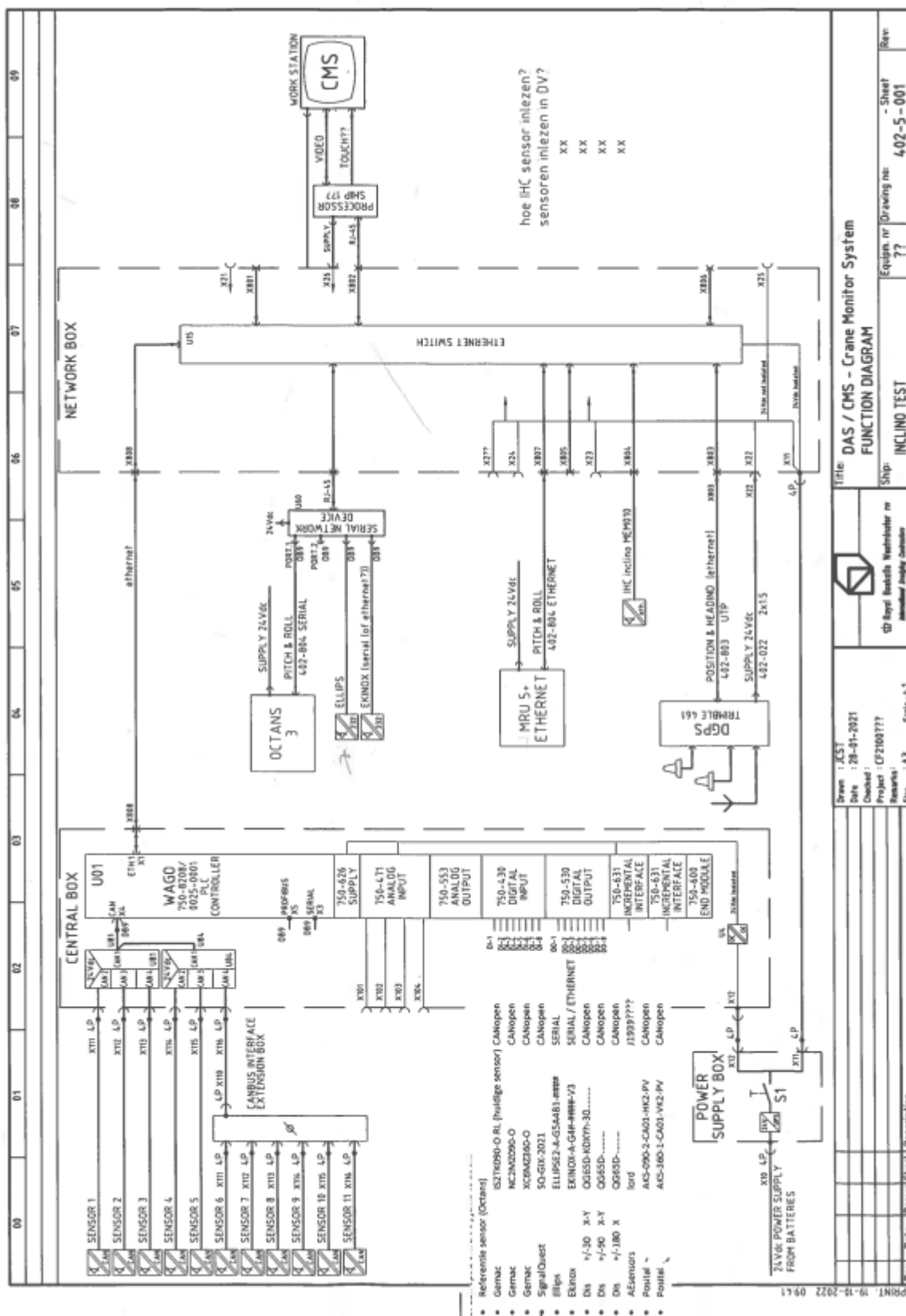
Y + Fore the MRU

X + SB of MRU

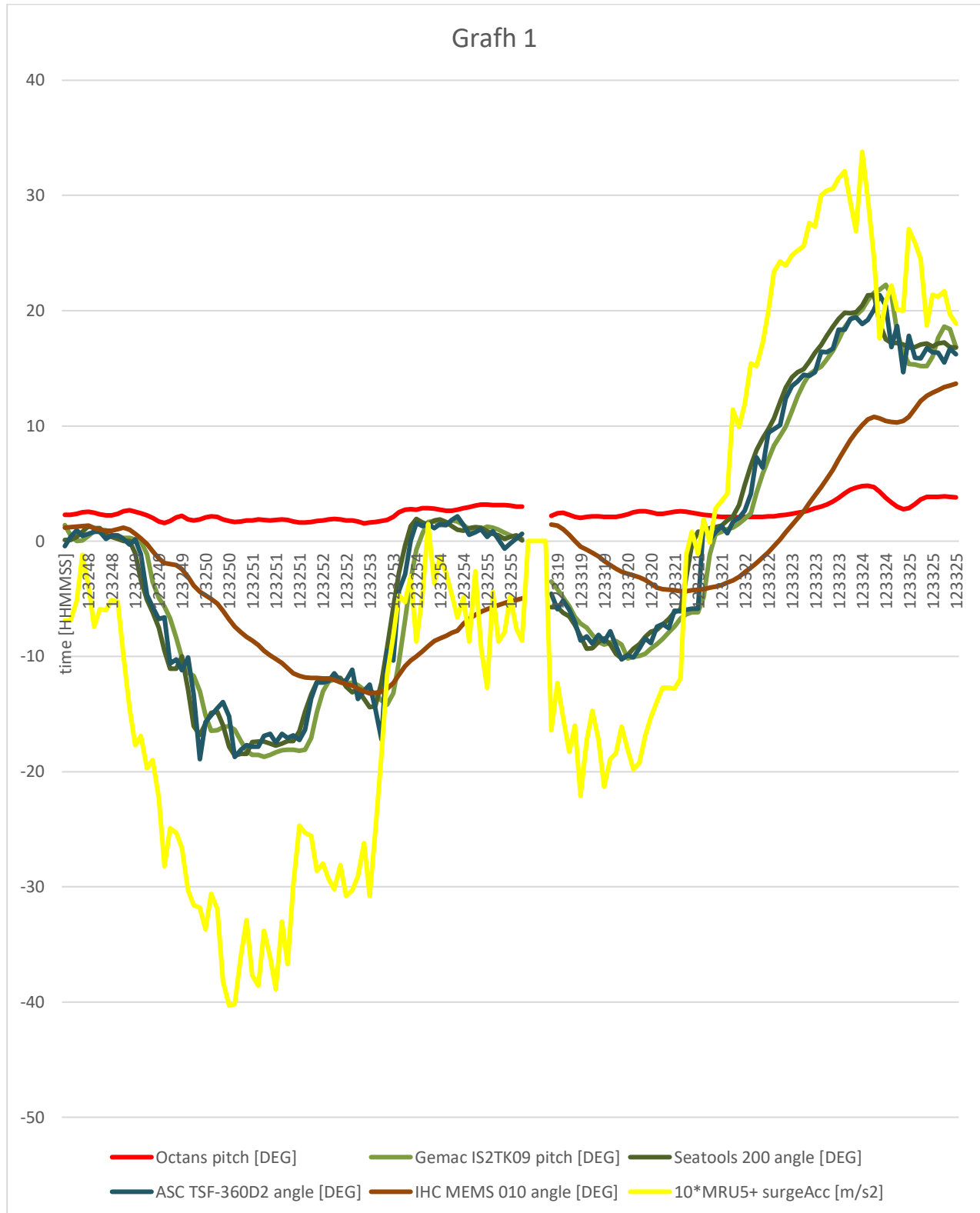
IXblue Octans 3	-Y -40mm	-X 318mm	-
SBG Ekinox	-Y -115mm	-X 433mm	-
SBG Ellipse	-Y 70mm	-X 443mm	-
Gemac IS2TK090-0-RL	-Y 60mm	-X -231mm	-
Gemac Motus XC6MZ360-O	-Y -25mm	-X -171mm	-
SignalQuest SQ-GIX-2021	-Y -120mm	-X -222mm	-
Posital AKS-090-2-CA01-HK2-PV	-Y 55mm	-X -101mm	-
Posital AKS-360-1-CA01-VK2-PV	-Y -410mm	-X -171mm	-
Dis QG65D-KDXYh-090H-Can-CFM-UL	-Y -260mm	-X -246mm	-
Lord sensing MV5-AR-J1939	-Y -300mm	-X 38mm	-
ASC TSF-360-CAN-D2	-Y -280mm	-X -202mm	-
IHC MEMS 010	-Y -650mm	-X -237mm	-
Seatools Sinclino 200	-Y -690mm	-X -107mm	-

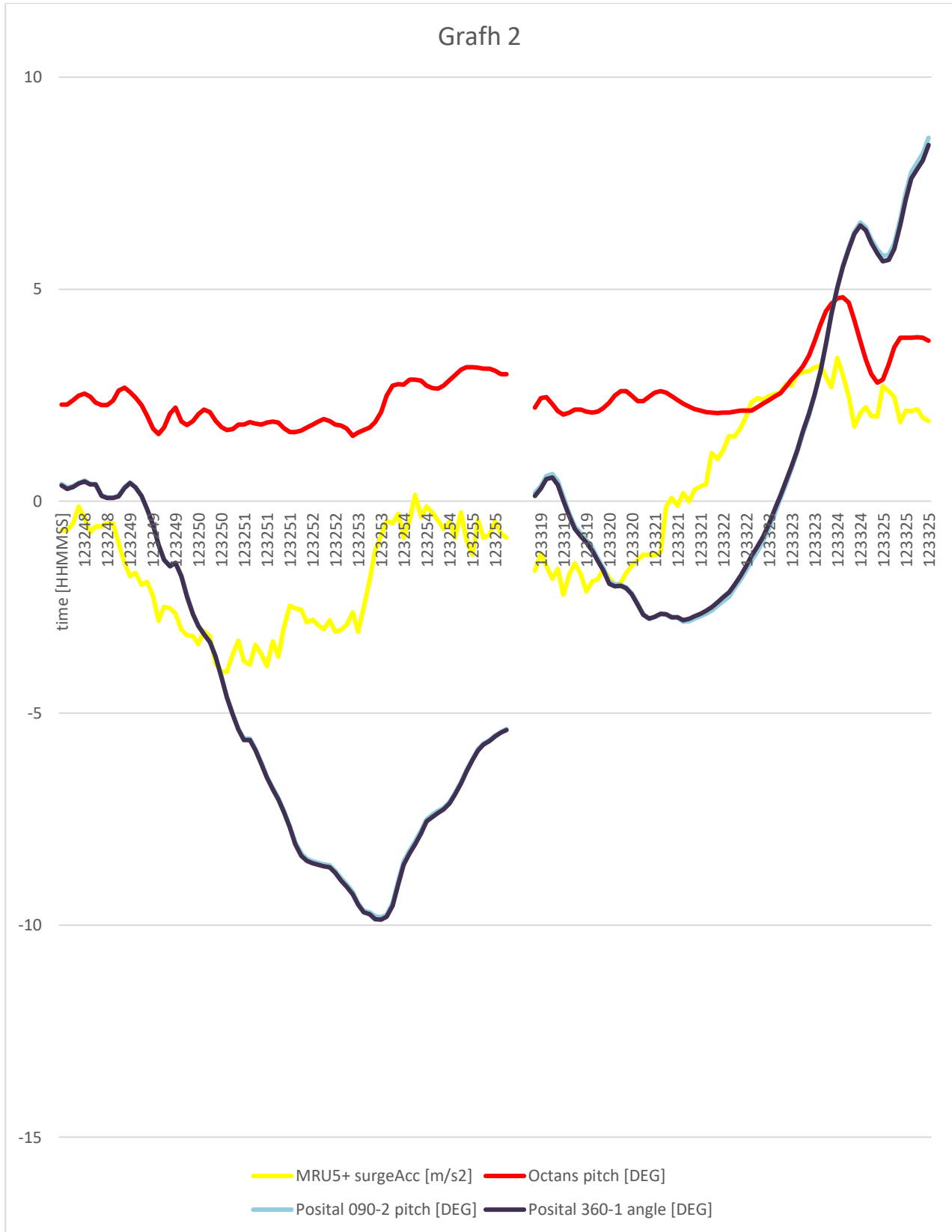
At Chapter 11 is a top view of the base

10. DRAWINGS

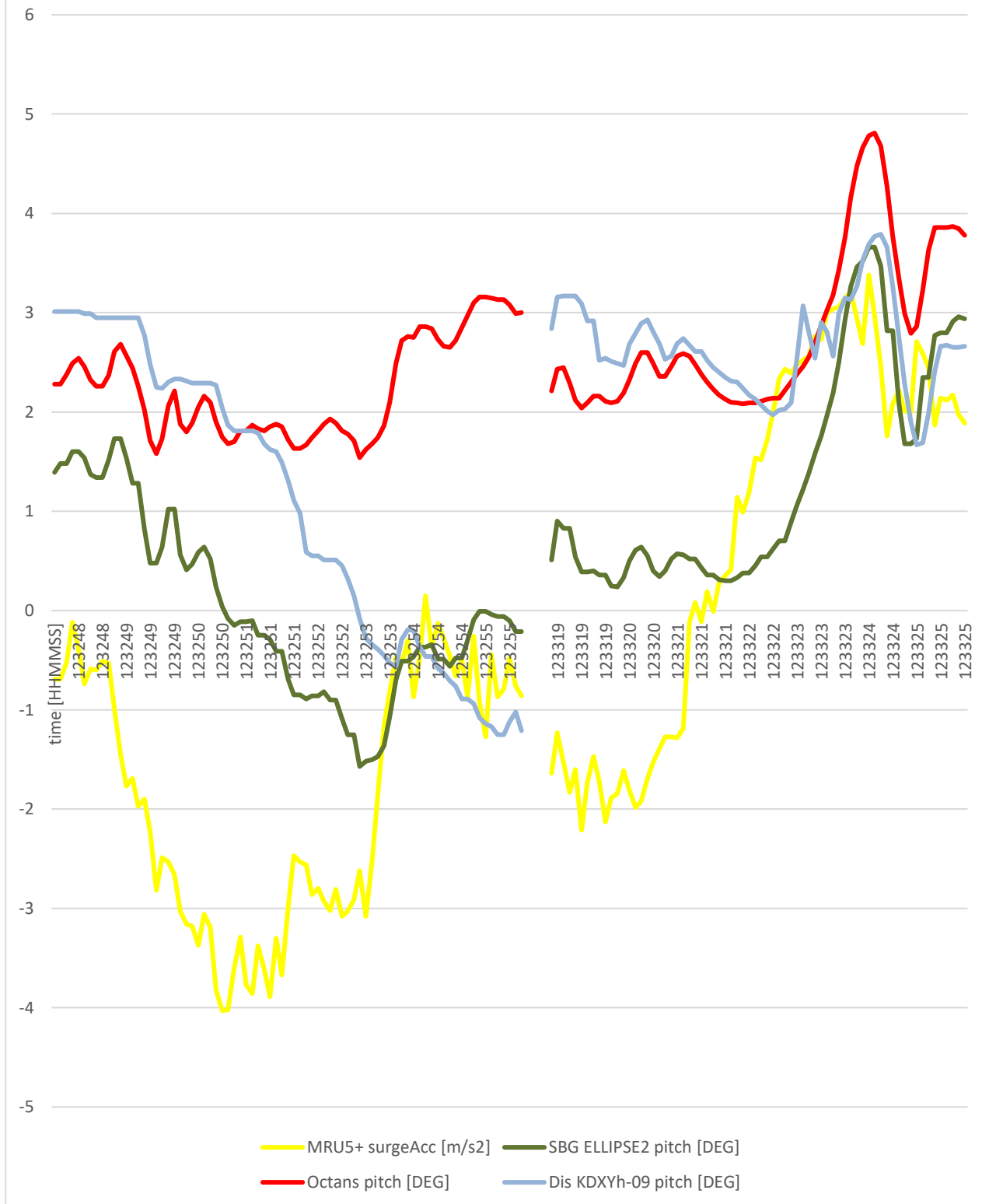


11. PICTURES

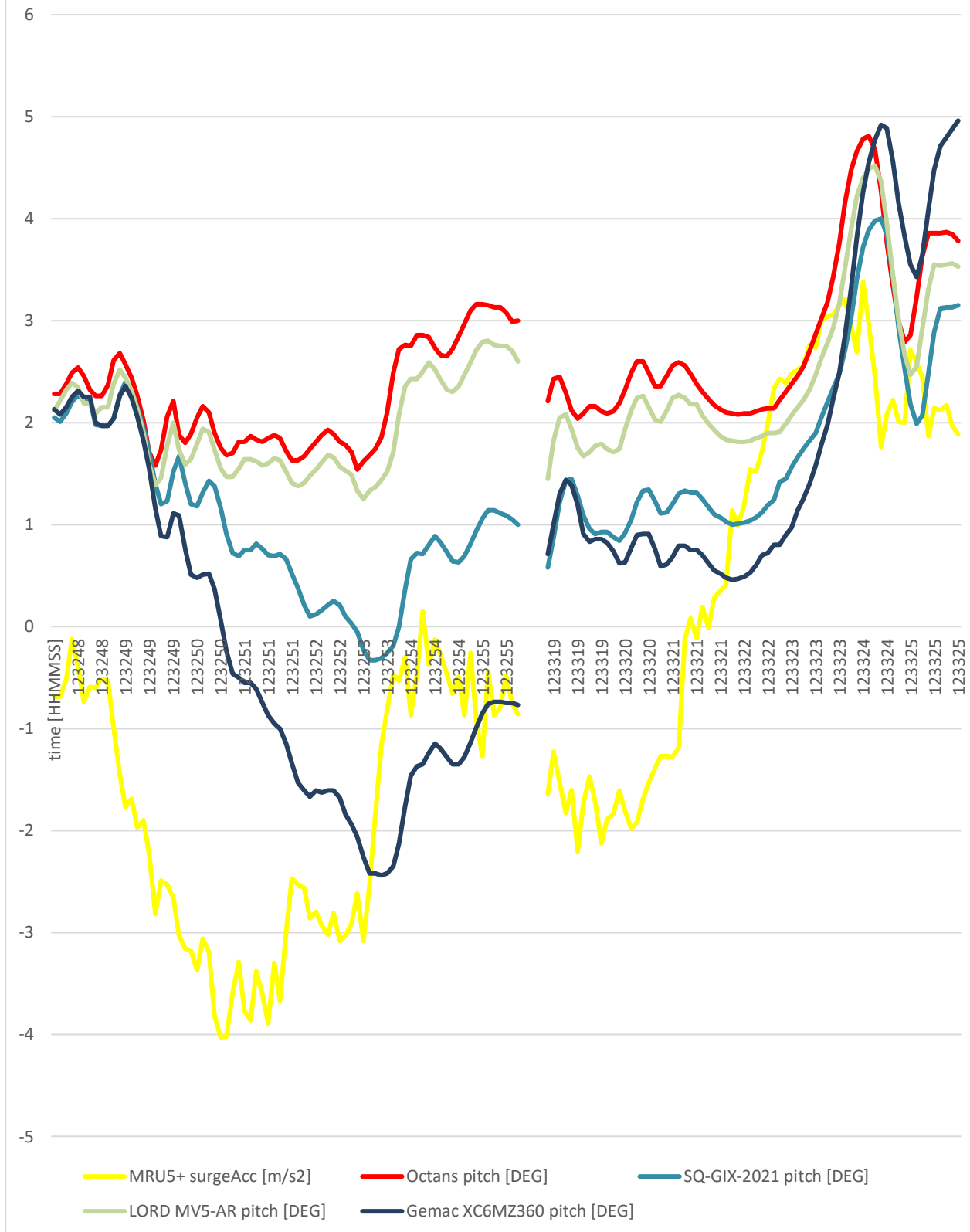


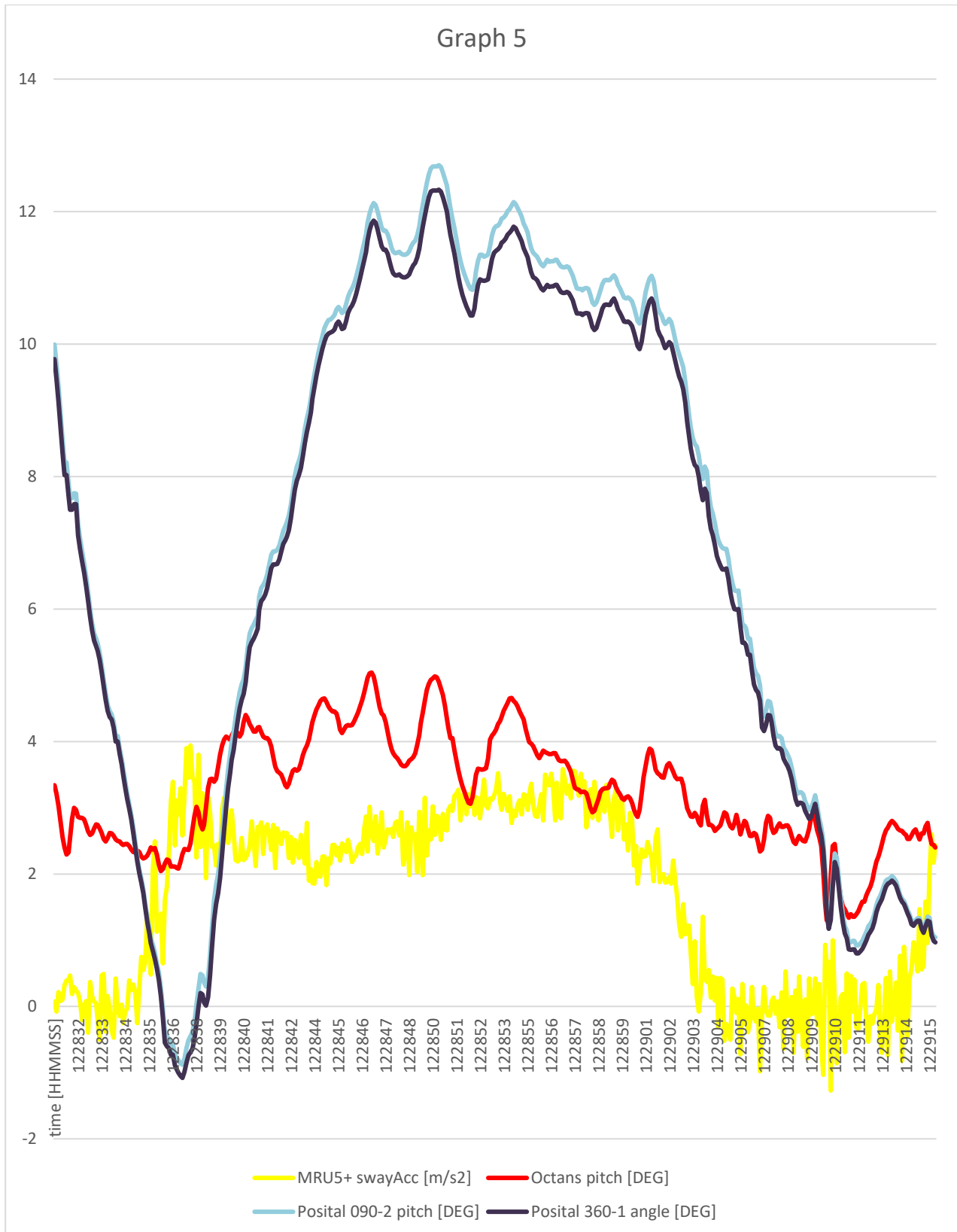


Graph 3

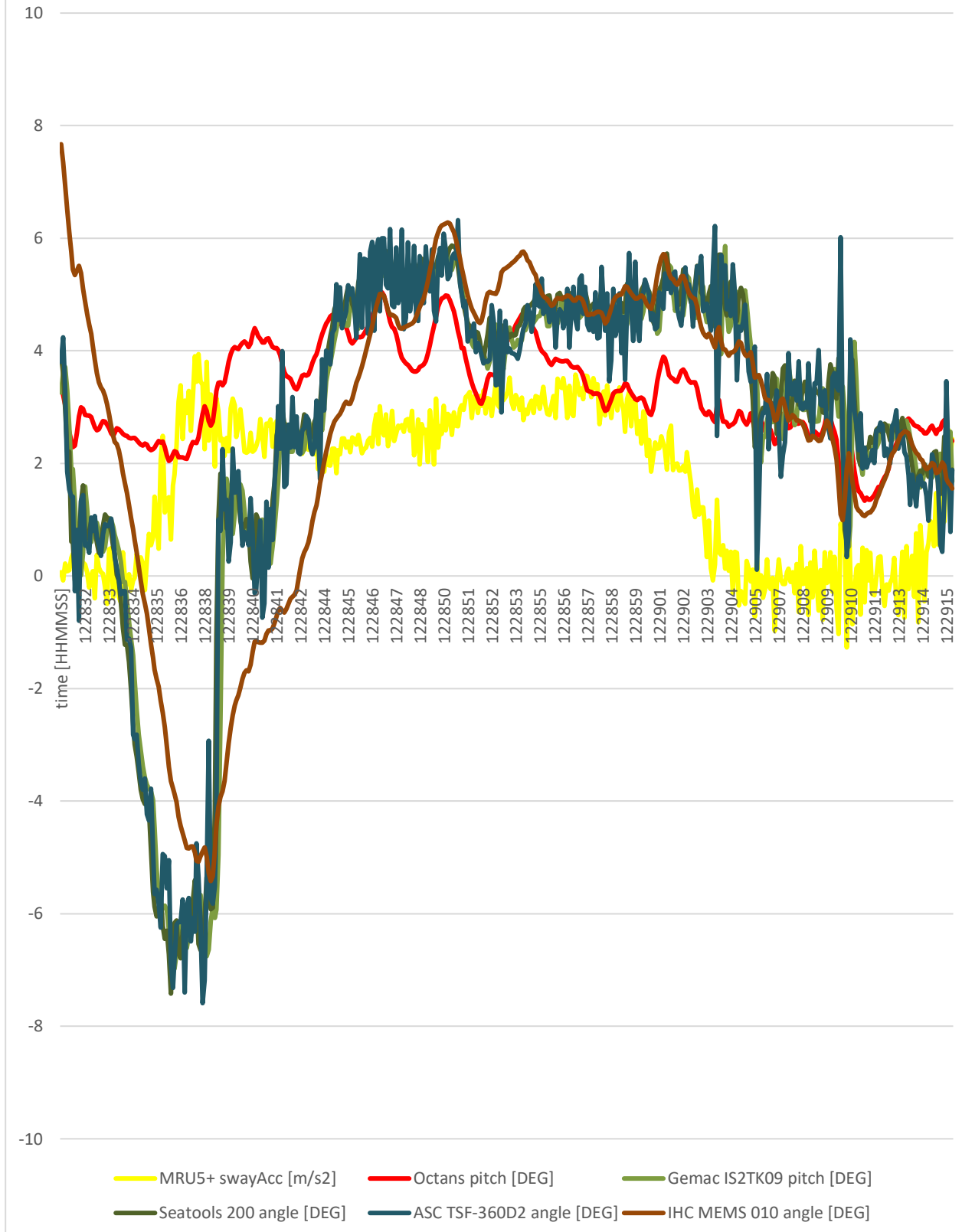


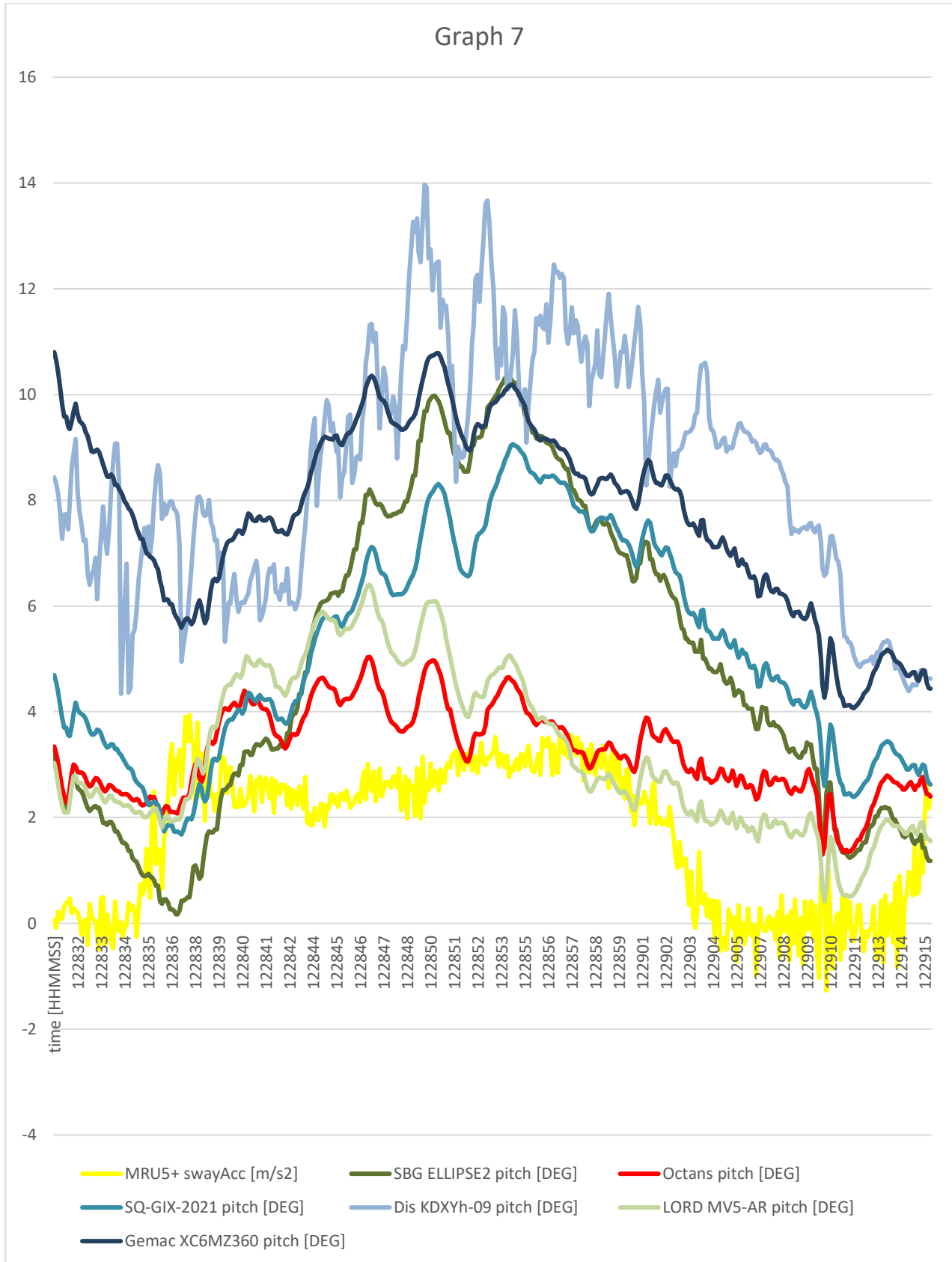
Graph 4

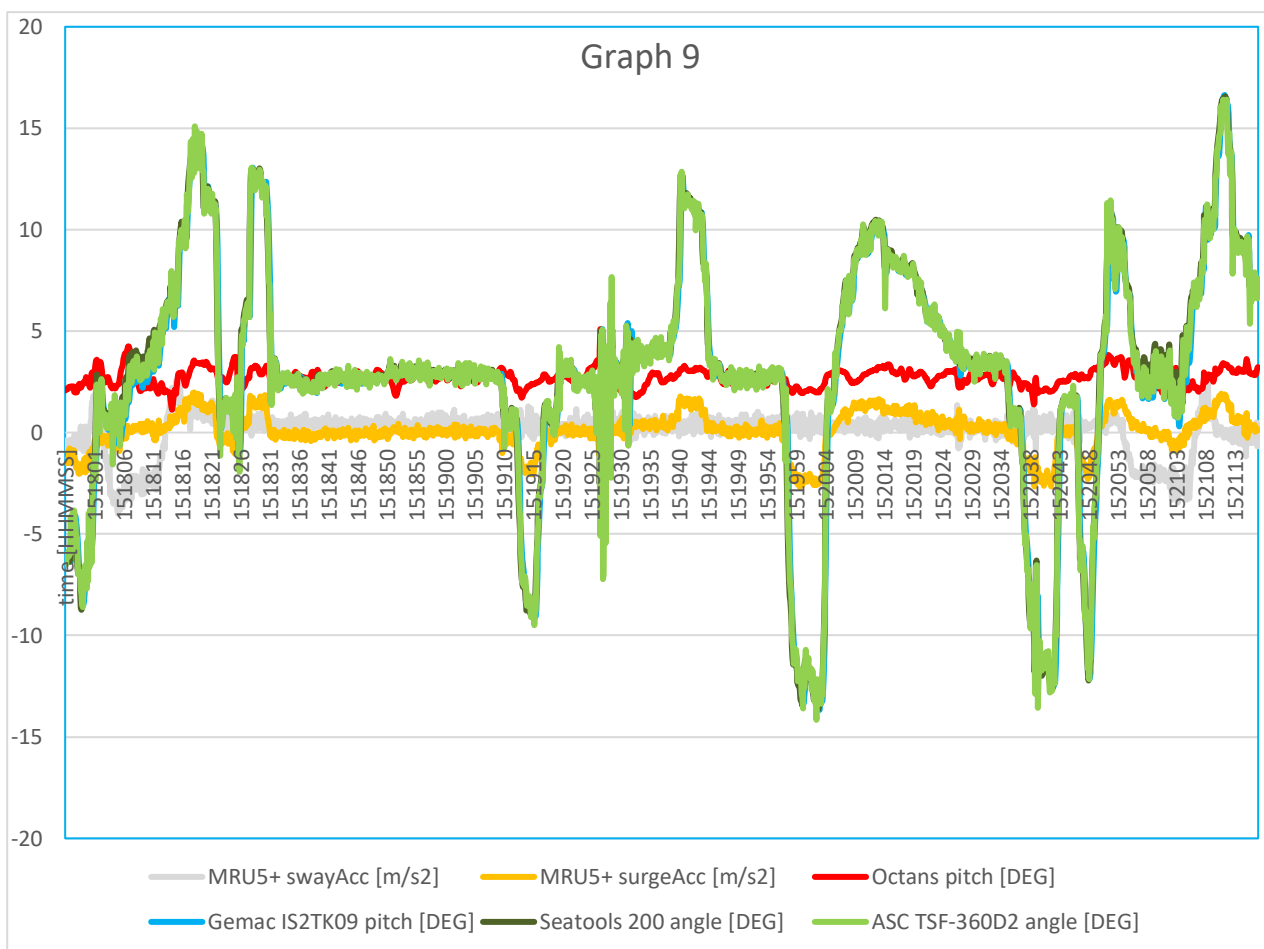
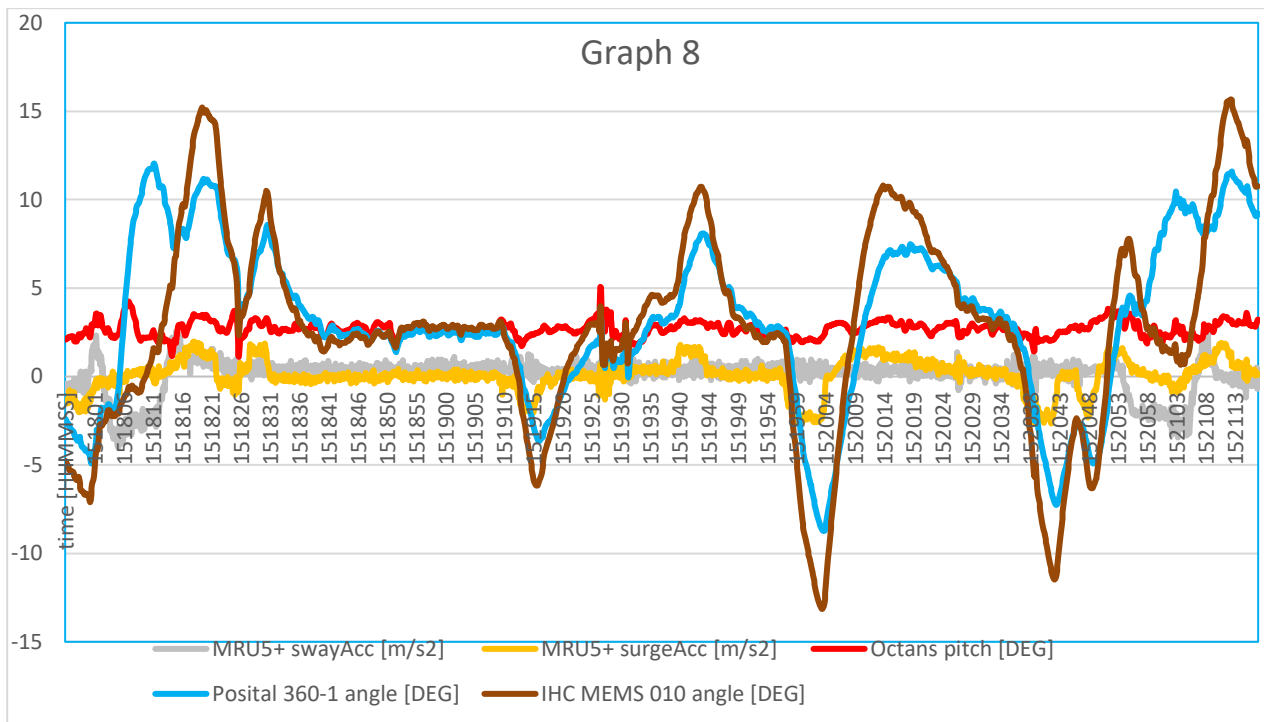


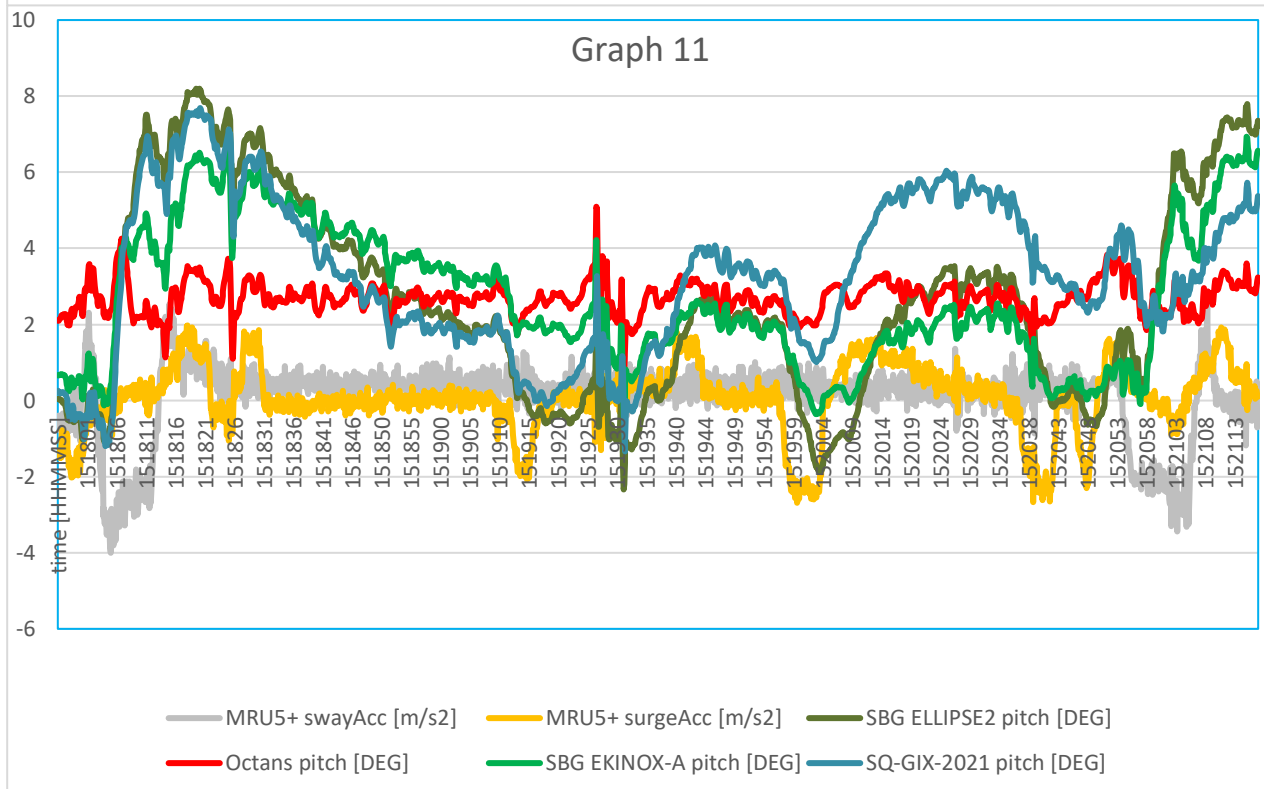
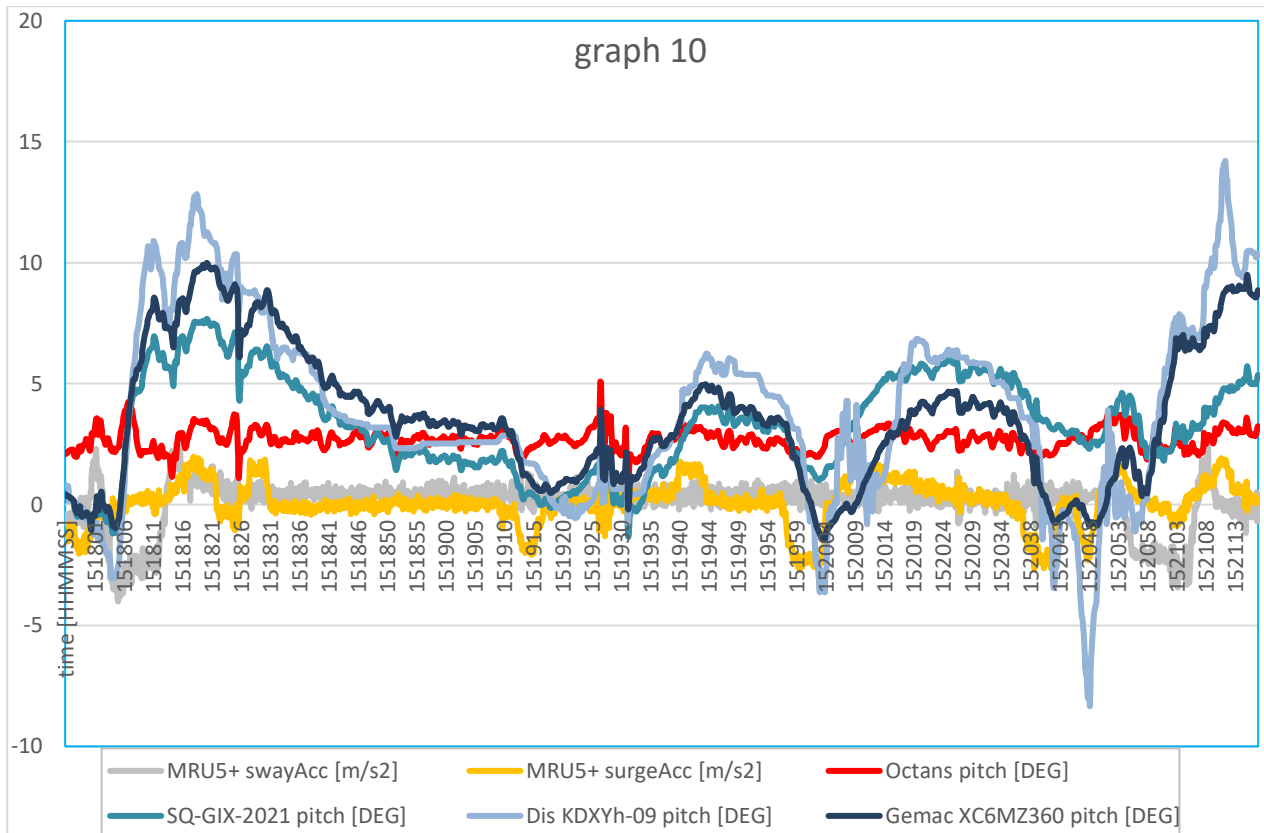


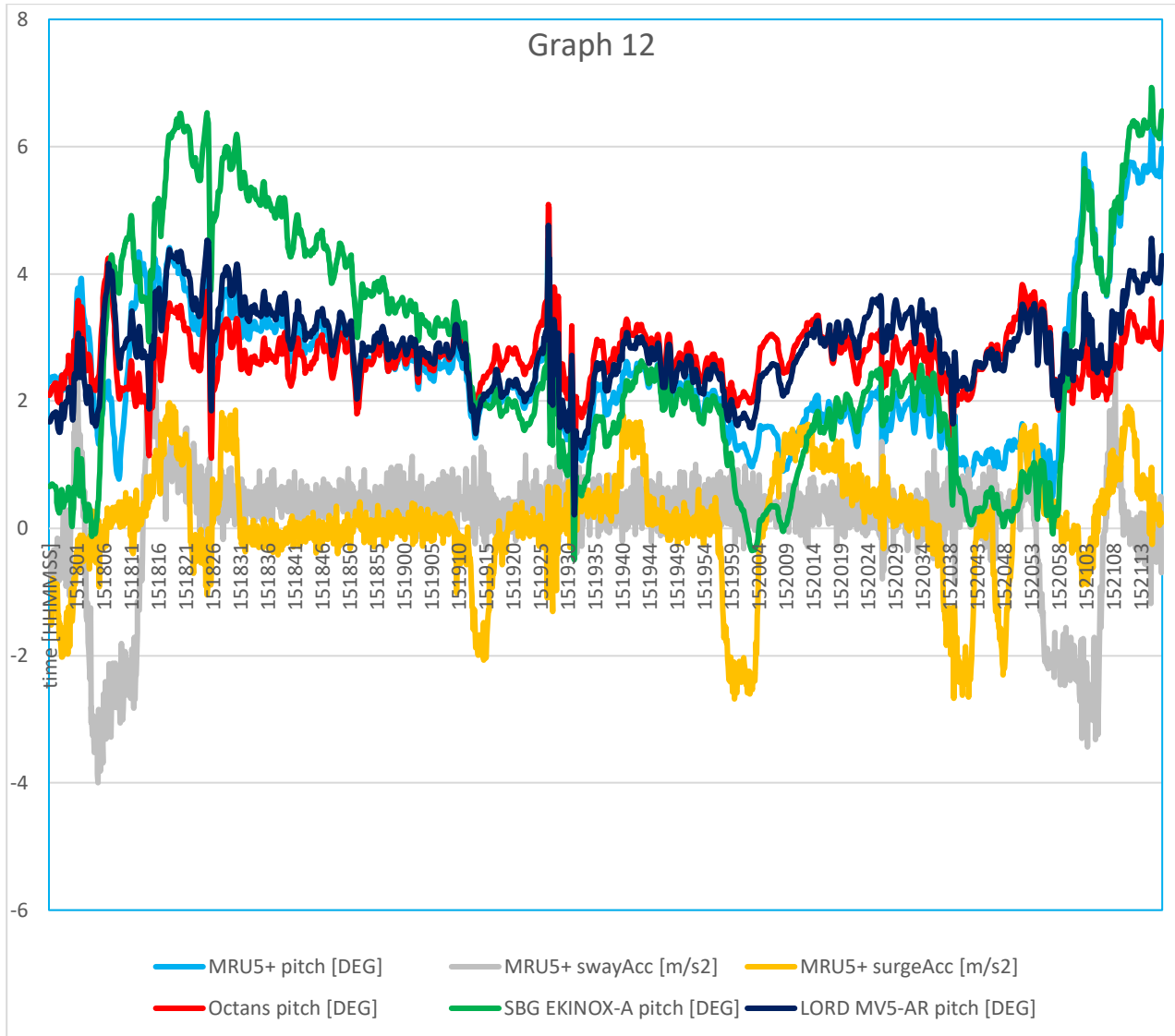
Graph 6



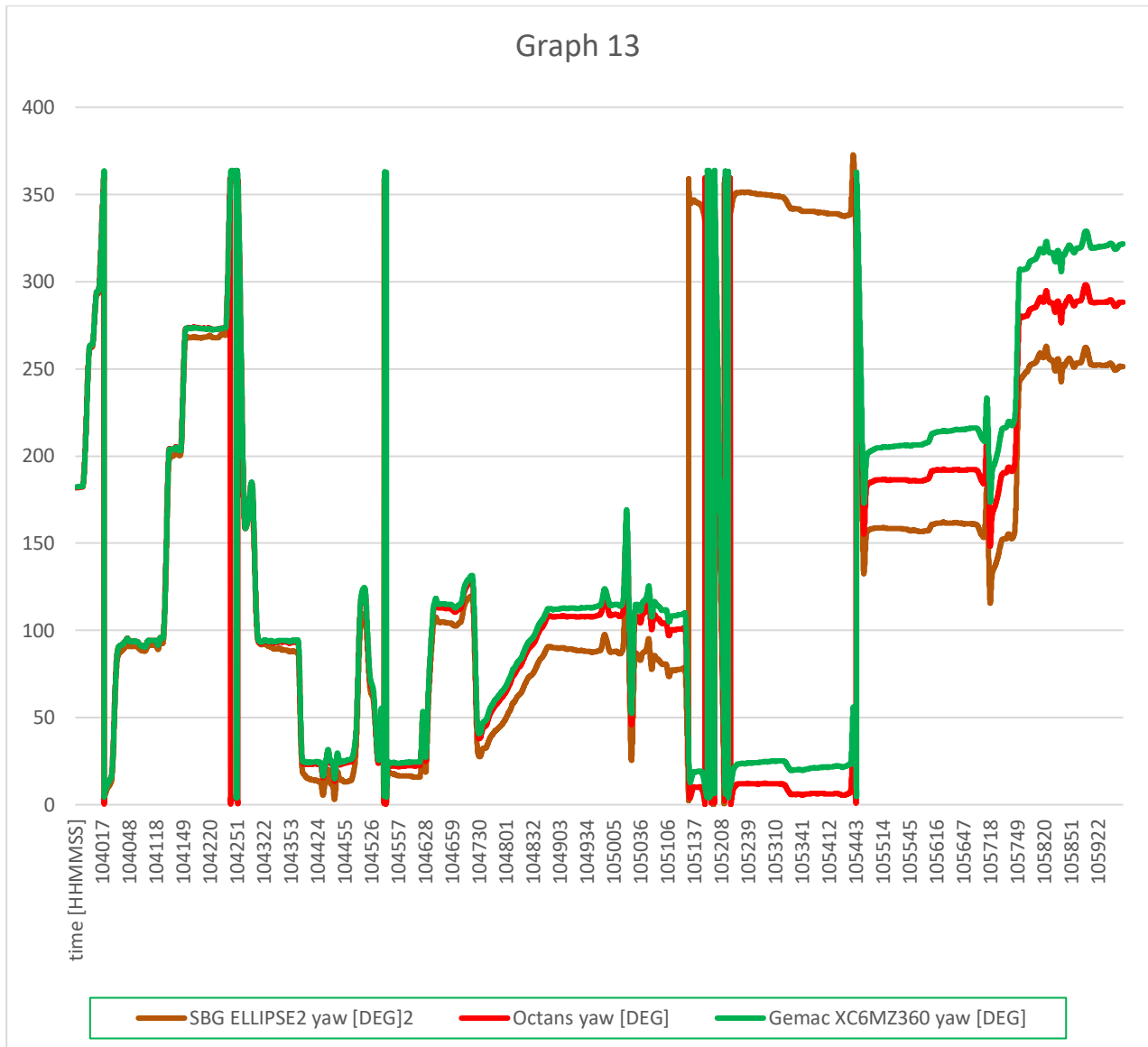




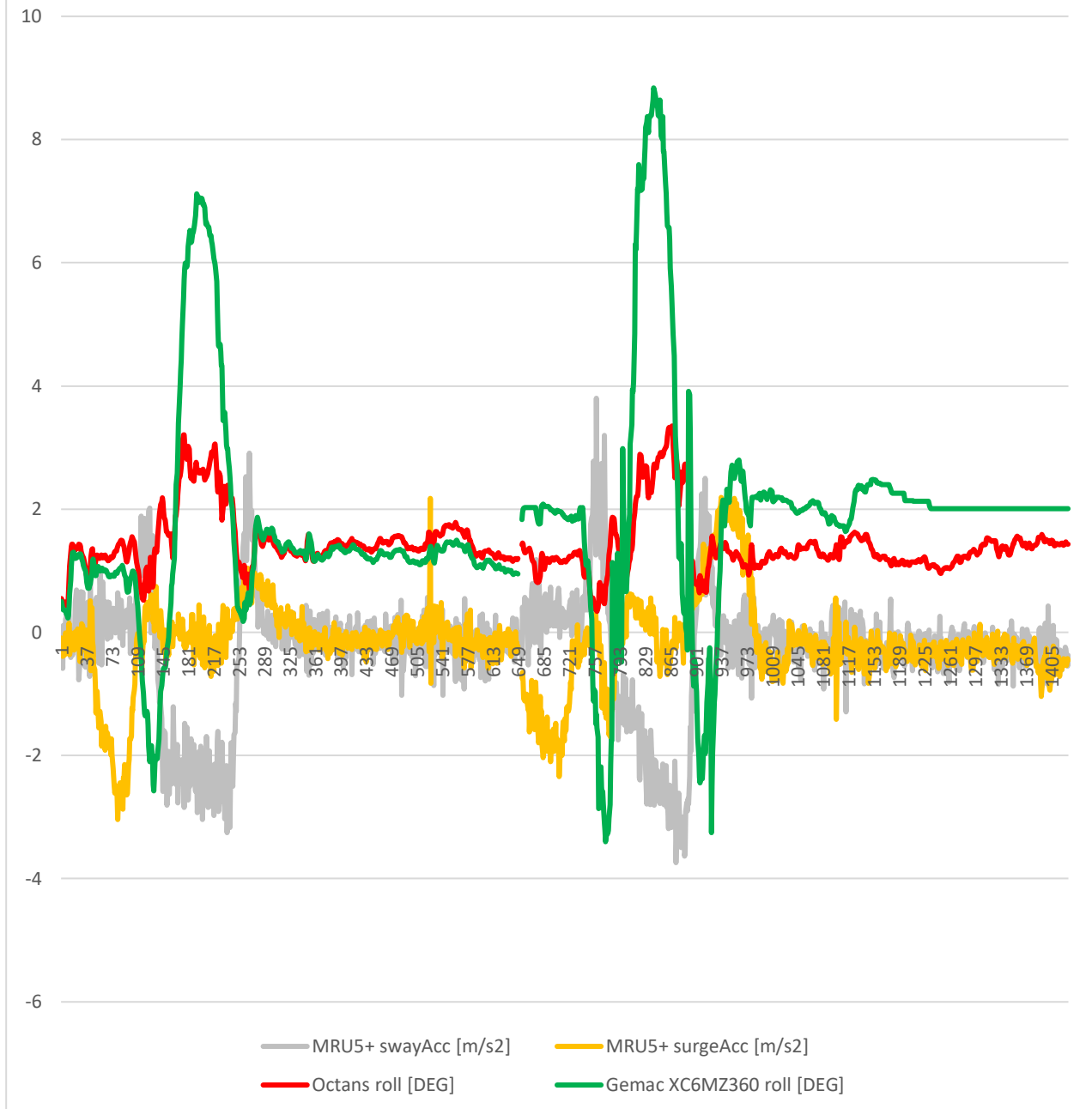




Graph 13

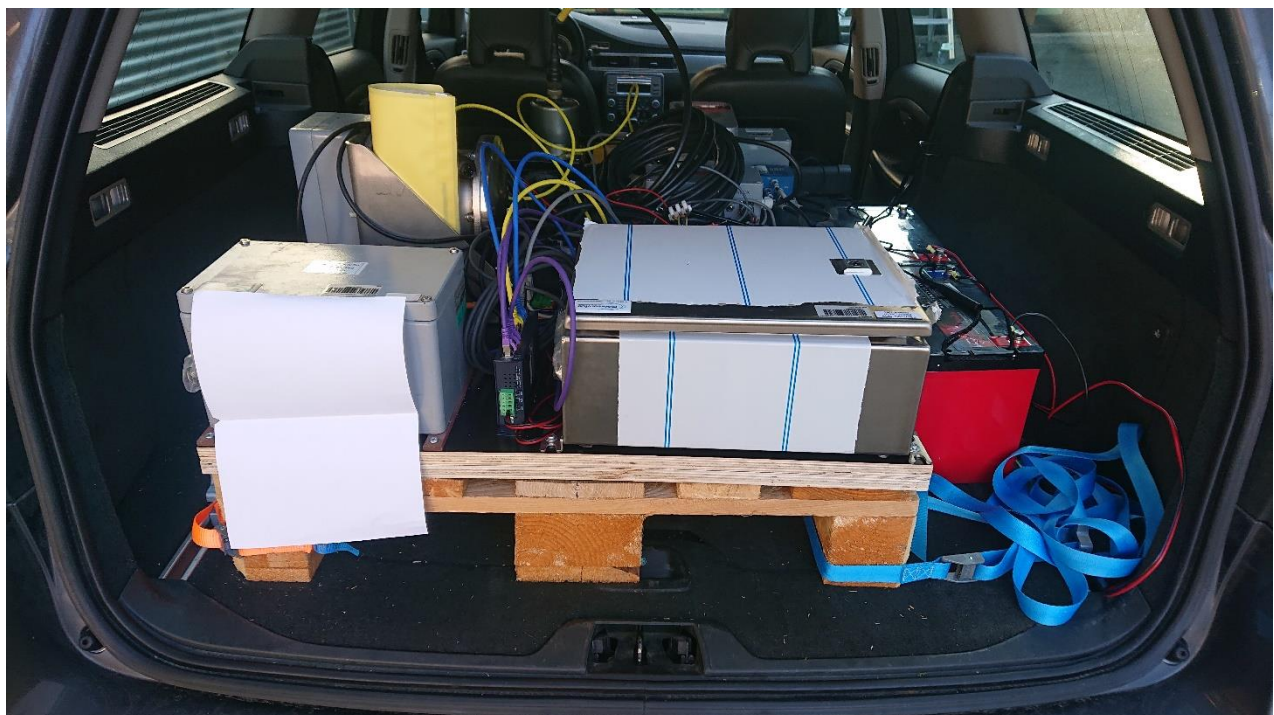


Graph 14

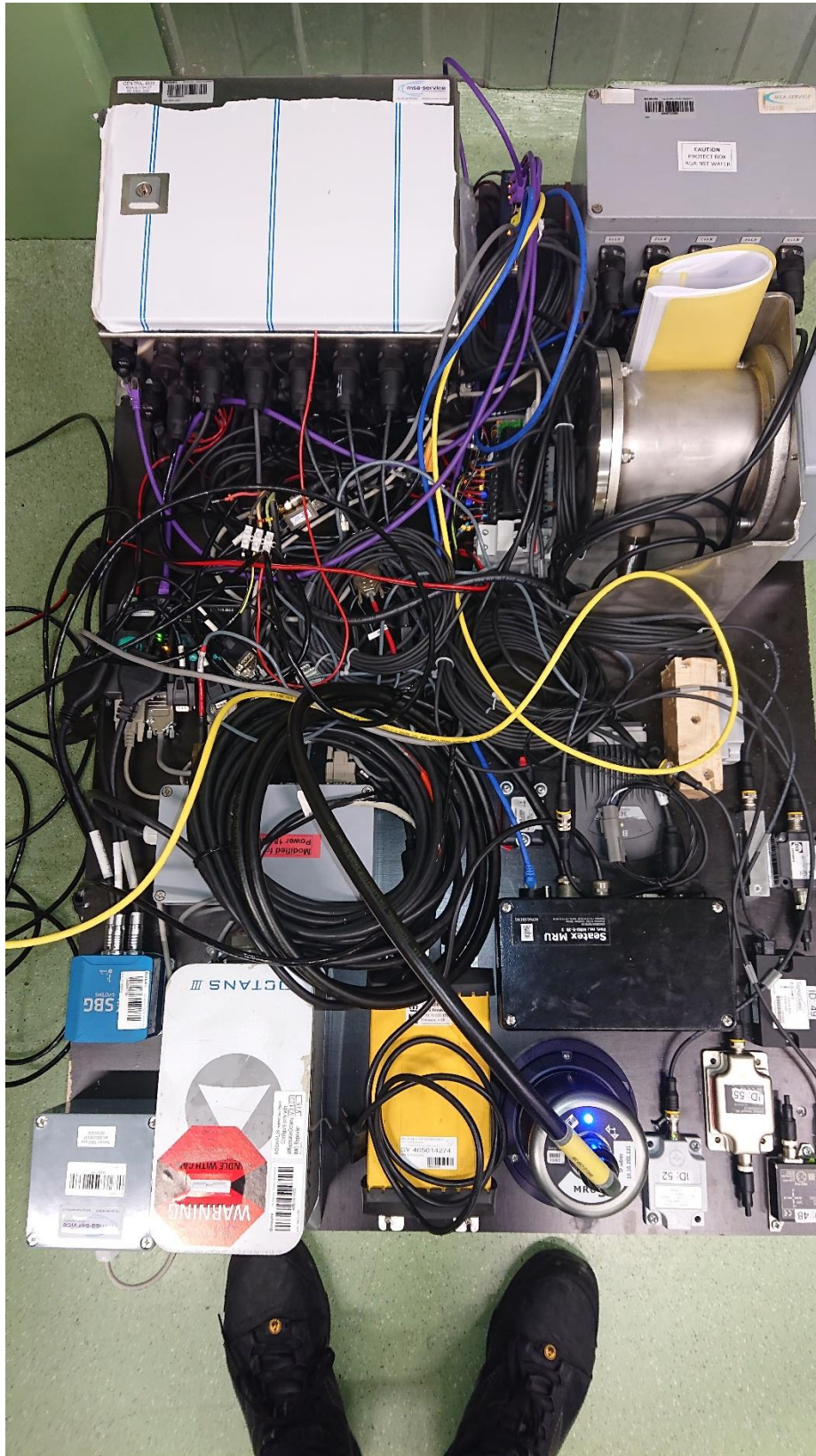




picture 1 Front side of base in the Car



picture 2 Base backside in Car



picture 3 Top view base